

Random Variables and Distributions

Numerical uncertainty, distributions, model families, simulation, and system contracts.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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1. START WITH THE VARIABLE CONTRACT

State the experiment, numerical mapping, unit, horizon, observation clock, and eligible population. Distribution work is invalid when the variable itself is ambiguous.

2. INSPECT SUPPORT BEFORE SHAPE

Ask whether values can be negative, bounded, discrete, zero-inflated, censored, or truncated. This eliminates many attractive but impossible model families.

3. SEPARATE DATA FROM MODEL

The sample is finite evidence; the distribution is a model of uncertainty. Compare empirical behavior with several defensible families and retain sampling uncertainty.

4. READ THE WHOLE DISTRIBUTION

Use PMF or density for local shape, CDF and survival for thresholds, quantiles for inverse questions, and moments only as summaries. Tails and modes can contradict a simple mean-variance story.

5. VALIDATE FOR THE DECISION

A model used for central forecasts may fail for loss limits. Evaluate support, held-out scores, PIT or Q-Q behavior, regime stability, and the exact tail region consumed downstream.

6. PRESERVE SYSTEM CONTEXT

Register transformations, parameter conventions, data lineage, conditioning variables, seeds, and supported operations. Every scenario or threshold should be replayable.

DECISION STANDARD

Promote a distribution only when its assumptions are explicit, its relevant region is validated out of sample, and a simpler empirical or parametric challenger has been considered.

Distribution notation and reading discipline

BEGINNER

SYMBOL	TYPE	MEANING
X, Y	random variables	Uncertain numerical outcome maps
x, y	realizations	Observed or candidate values
$\text{support}(X)$	domain	Values allowed by the model
$p_X(x)$	PMF	Point probability for discrete X
$f_X(x)$	PDF	Density per unit for continuous X
$F_X(x)$	CDF	Probability X is at most x
$S_X(x)$	survival	Probability X exceeds x
$Q_X(p)$	quantile	Threshold at cumulative probability p
μ, σ	moments	Mean and standard deviation
θ	parameters	Family-specific fitted quantities

UNITS AND PARAMETERS

A scale parameter inherits units from X , variance uses squared units, and density uses inverse units. Parameter names are not portable across libraries: gamma may use rate or scale, and negative-binomial conventions vary.

PROBABILITY INTERFACE

Use PMF for discrete point questions, CDF or survival for thresholds, quantile for inverse thresholds, and log-PMF or log-PDF for likelihood calculations. Do not approximate a tail probability from density height.

Distribution modeling workflow

BEGINNER

1

Define X

Outcome map, unit, horizon, clock, population

2

Audit support

Bounds, zeros, discreteness, censoring, missingness

3

Explore shape

ECDF, histogram, tails, regimes, dependence

4

Choose challengers

Empirical baseline and plausible model families

5

Fit honestly

Training window, stable likelihood, uncertainty

6

Validate

Held-out scores, PIT/Q-Q, tails, stability

7

Publish contract

Parameters, transforms, operations, lineage

8

Monitor

Support drift, calibration, regime and tail breaches

HARD GATE

Do not publish probabilities or scenarios until support, units, conditioning, parameter convention, held-out behavior, and the decision-relevant tail have passed validation.

Random variables as outcome maps

BEGINNER

ONE-SENTENCE MEANING

A random variable assigns a numerical value to every modeled outcome. Its definition includes the measurement rule, unit, horizon, and information timing, not merely a column name.

MEMORY JOGGER

Think of X as a function from uncertain market outcomes to numbers. Once the map is fixed, probabilities on outcomes induce a distribution over X .

WHY IT MATTERS IN QUANT RESEARCH

Returns, P&L, spread changes, trade counts, fills, and drawdowns become analyzable only after their random-variable contracts are explicit. This prevents models from silently mixing incompatible quantities.

FORMULA INTUITION

$X: \Omega \rightarrow \mathbb{R}; P(X \text{ in } A) = P(\{\omega: X(\omega) \text{ in } A\})$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Let X be tomorrow's close-to-close return after corporate actions. A +1.2% day maps to 0.012, while an unavailable close maps to a declared missing state rather than zero.

PYTHON / SYSTEM IMPLEMENTATION

Store variable name, unit, calendar, timestamp convention, horizon, transformation, valid domain, and missing policy in the schema registry. Add tests that recompute a small reference sample from raw data.

CONFUSIONS TO AVOID

Calling any numeric feature a random variable without defining its experiment hides selection and timing assumptions. Prices, returns, and P&L also have different supports and aggregation rules.

WHERE THIS FITS IN THE RESEARCH SYSTEM

This contract sits between cleaned observations and every distribution fit, simulation, forecast, or risk metric. Downstream artifacts should reference the contract version.

RELATED CONCEPTS AND QUICK RECALL

Related: sample space, measurable function, data contract. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Realizations, samples, and populations

BEGINNER

ONE-SENTENCE MEANING

A realization is one observed value of a random variable, while a sample is a collection of realizations from a specified population or process. The population is conceptual and may change through time.

MEMORY JOGGER

Uppercase X denotes the uncertain quantity; lowercase x denotes a realized value. A dataset is evidence about a distribution, not the distribution itself.

WHY IT MATTERS IN QUANT RESEARCH

Quant research estimates distributional behavior from finite observations and must state which market, assets, dates, and eligibility rules generated them. Without that population definition, estimates cannot be interpreted or reproduced.

FORMULA INTUITION

$X_1, \dots, X_n$ are random variables; $x_1, \dots, x_n$ are observed realizations. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A sample of 1,250 daily returns from current index members is not evidence about all historical constituents. Survivorship changes the population before any statistic is calculated.

PYTHON / SYSTEM IMPLEMENTATION

Persist the sampling frame, as-of dates, membership history, exclusions, and row-level lineage. Keep train, validation, and test samples immutable once an experiment is registered.

CONFUSIONS TO AVOID

Treating rows as independent copies is often false because returns share dates, factors, and regimes. More rows can therefore add far less information than the count suggests.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The sampling frame is the bridge from data engineering to inference. Model validation should reject results whose claimed population differs from the actual extraction logic.

RELATED CONCEPTS AND QUICK RECALL

Related: realization, sampling frame, effective sample size. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Discrete and continuous variables

BEGINNER

ONE-SENTENCE MEANING

A discrete variable takes values from a countable support and assigns probability mass to points. A continuous model assigns density across intervals, so individual points normally have probability zero.

MEMORY JOGGER

Discrete means sum masses; continuous means integrate density. Observed tick-size granularity does not automatically make a continuous approximation inappropriate.

WHY IT MATTERS IN QUANT RESEARCH

The distinction determines likelihoods, simulation methods, visualizations, and whether zero or boundary outcomes require explicit mass. It also guides how fills, counts, returns, and durations should be modeled.

FORMULA INTUITION

Discrete: $\sum_x p(x)=1$; continuous: $\int f(x) dx=1$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Order count in a minute is discrete, while execution delay is often modeled continuously. Daily return is recorded to finite precision but is commonly treated as continuous for inference.

PYTHON / SYSTEM IMPLEMENTATION

Preserve integer and categorical types for discrete outcomes. When binning a continuous variable, store the original values and exact bin edges so the transformation remains reversible.

CONFUSIONS TO AVOID

A density value is not a point probability and may exceed one when units are narrow. Conversely, forcing count data into a continuous model can assign mass to impossible negative or fractional values.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Support type belongs in the semantic schema because fitting, scoring, and sampling services branch on it. Validation should compare simulated support with the declared domain.

RELATED CONCEPTS AND QUICK RECALL

Related: PMF, PDF, support. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Support and boundary behavior

BEGINNER

ONE-SENTENCE MEANING

The support is the set of values a random variable can take under a model. Boundaries may be finite, infinite, open, closed, absorbing, or associated with separate point mass.

MEMORY JOGGER

Check what values are possible before choosing a distribution. A model that leaks probability outside the feasible domain is structurally wrong even if its center fits well.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio weights, volatilities, counts, probabilities, and losses each impose different constraints. Correct support prevents impossible simulations and unstable likelihood calculations near limits.

FORMULA INTUITION

$\text{support}(X) = \{x: p(x) > 0\}$ or $\text{closure}\{x: f(x) > 0\}$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A Gaussian model for bid-ask spread can generate negative spreads. A lognormal or gamma alternative respects positivity but still needs validation near zero and in the upper tail.

PYTHON / SYSTEM IMPLEMENTATION

Encode lower and upper bounds, inclusivity, point masses, and transformation rules. Unit tests should sample large batches and fail if generated values violate the contract.

CONFUSIONS TO AVOID

Clipping impossible draws after simulation changes the distribution and usually creates artificial mass at the boundary. Use a model with the right support or a principled truncated distribution.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Support checks are hard gates in the distribution registry and scenario engine. They should run before goodness-of-fit rankings because no statistical score rescues impossible outcomes.

RELATED CONCEPTS AND QUICK RECALL

Related: truncation, censoring, transformation. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Probability mass functions

BEGINNER

ONE-SENTENCE MEANING

A probability mass function gives the probability of each possible value of a discrete random variable. All masses are nonnegative and sum to one across the complete support.

MEMORY JOGGER

For a PMF, bar height is probability rather than density. Missing support values still matter when they are possible but simply unobserved in a sample.

WHY IT MATTERS IN QUANT RESEARCH

PMFs describe return signs, fill states, signal votes, outages, defaults, and trade counts with small supports. They also provide exact likelihoods for discrete models.

FORMULA INTUITION

$p_X(x) = P(X=x)$; $p_X(x) \geq 0$; $\sum_x p_X(x) = 1$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Suppose fill count X has masses $P(0)=0.20$, $P(1)=0.50$, $P(2)=0.25$, $P(3)=0.05$. The chance of at least two fills is 0.30.

PYTHON / SYSTEM IMPLEMENTATION

Represent support and probability arrays in the same deterministic order. Validate duplicates, normalization tolerance, and whether every schema state appears even when estimated mass is zero.

CONFUSIONS TO AVOID

Dropping unseen states and renormalizing makes the empirical PMF overconfident. Pooling rare states without a documented business meaning can also destroy useful tail information.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A discrete distribution object feeds exact scenario weights, likelihood scoring, and categorical sampling. Version the support separately from estimated masses so ontology changes are visible.

RELATED CONCEPTS AND QUICK RECALL

Related: categorical distribution, histogram, empirical frequency. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Probability density functions

BEGINNER

ONE-SENTENCE MEANING

A probability density function allocates probability per unit of a continuous variable. Probability is area under the density across an interval, not the height at one point.

MEMORY JOGGER

Density answers where probability is concentrated locally. The vertical scale changes when units change, but integrated probability does not.

WHY IT MATTERS IN QUANT RESEARCH

Densities support return likelihoods, continuous scenario generation, risk thresholds, and comparison of model families. They expose skew, tail thickness, and multimodality more clearly than a few moments.

FORMULA INTUITION

$P(a \leq X \leq b) = \int_a^b f_X(x) dx$; $\int f_X(x) dx = 1$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If a daily-return density is high near zero, that does not mean $P(X=0)$ is high. The relevant chance is the area in a band such as -0.1% to $+0.1\%$.

PYTHON / SYSTEM IMPLEMENTATION

Use stable library PDF and log-PDF functions, preserve units, and verify numerical integration on test parameters. Compare analytic probabilities with Monte Carlo frequencies in fixed intervals.

CONFUSIONS TO AVOID

Reading $f(x)$ as $P(X=x)$ produces nonsense for continuous outcomes. Comparing densities measured in decimals and percentage points without a Jacobian also creates false discrepancies.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The fitted density and its transformation metadata belong together in the model registry. Risk services should request CDF or survival probabilities instead of approximating area from height.

RELATED CONCEPTS AND QUICK RECALL

Related: CDF, log-likelihood, Jacobian. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Cumulative distribution functions

BEGINNER

ONE-SENTENCE MEANING

The cumulative distribution function gives the probability that a random variable is at or below a threshold. Every valid CDF is nondecreasing, right-continuous, and approaches zero and one at the extremes.

MEMORY JOGGER

The CDF converts a value into cumulative probability. It works for discrete, continuous, and mixed distributions with the same interpretation.

WHY IT MATTERS IN QUANT RESEARCH

CDFs answer loss-limit probabilities, percentile ranks, and probability-integral-transform diagnostics. They are often the safest common interface across distribution families.

FORMULA INTUITION

$F_X(x) = P(X \leq x)$; $P(a < X \leq b) = F(b) - F(a)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If $F(-0.02) = 0.04$, the model assigns a 4% chance to returns of -2% or worse. If $F(0.01) = 0.72$, then 28% of mass lies above +1%.

PYTHON / SYSTEM IMPLEMENTATION

Expose vectorized `cdf`, `logcdf`, `sf`, and `ppf` methods with parameter validation. Test monotonicity and compare interval probabilities against direct numerical integration or exact sums.

CONFUSIONS TO AVOID

Subtracting two nearly equal CDF values in an extreme upper tail can lose precision. Use the survival function or log-survival implementation when available.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The CDF interface connects fitted distributions to thresholds, backtest diagnostics, calibration tests, and scenario quantiles. It should be standardized across the model library.

RELATED CONCEPTS AND QUICK RECALL

Related: survival function, quantile, PIT. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Survival functions and hazard rates

BEGINNER

ONE-SENTENCE MEANING

The survival function gives probability above a threshold, while the hazard rate measures instantaneous event intensity conditional on survival to that point. Together they describe tail exceedance and timing risk.

MEMORY JOGGER

Survival asks what remains beyond x . Hazard asks how quickly failure arrives now, given it has not arrived earlier.

WHY IT MATTERS IN QUANT RESEARCH

These functions are useful for order lifetimes, time-to-fill, outage duration, default timing, and tail-loss exceedances. They distinguish memoryless behavior from aging or duration dependence.

FORMULA INTUITION

$S(x)=P(X>x)=1-F(x)$; $h(x)=f(x)/S(x)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

For exponential fill time with rate 0.5 per second, $S(4)=\exp(-2)=13.5\%$. Its hazard stays 0.5, so waiting longer does not change the instantaneous modeled fill rate.

PYTHON / SYSTEM IMPLEMENTATION

Call survival and log-survival functions directly in tails. For duration data, retain censored observations and validate hazard shape rather than fitting only completed events.

CONFUSIONS TO AVOID

Discarding unfilled orders biases duration estimates toward fast fills. Treating hazard as an unconditional probability also misstates its conditional, per-unit-time meaning.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Duration models sit beside execution and operational-risk services. Their outputs should retain origin time, censoring status, covariates, and time units.

RELATED CONCEPTS AND QUICK RECALL

Related: duration model, censoring, exponential distribution. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Quantiles and inverse CDFs

BEGINNER

ONE-SENTENCE MEANING

A quantile is a threshold whose CDF reaches a target probability. The inverse CDF maps a probability level to an outcome and may require a convention when the distribution has jumps.

MEMORY JOGGER

CDF maps value to probability; quantile maps probability to value. Quantiles preserve ordering and remain interpretable for skewed or heavy-tailed data.

WHY IT MATTERS IN QUANT RESEARCH

Quantiles define VaR-style thresholds, scenario buckets, winsorization cutoffs, and robust summaries. They also transform uniform random draws into samples from a target distribution.

FORMULA INTUITION

$Q(p) = \inf\{x: F(x) \geq p\}$; $F(Q(p)) \geq p$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If the 1% return quantile is -4.2%, only about 1% of modeled returns lie at or below that threshold. A discrete distribution may have several p values mapping to the same outcome.

PYTHON / SYSTEM IMPLEMENTATION

Use a documented quantile convention and stable ppf routine. Store probability level, horizon, unit, fit date, and interpolation method with every reported threshold.

CONFUSIONS TO AVOID

A quantile does not describe average loss beyond the threshold. Comparing quantiles from different horizons without scaling assumptions can also be seriously misleading.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Quantile services feed risk limits, scenario generation, feature clipping, and monitoring. Expected shortfall or tail means should be separate outputs rather than inferred from VaR alone.

RELATED CONCEPTS AND QUICK RECALL

Related: percentile, VaR, inverse transform. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Expected value from a distribution

BEGINNER

ONE-SENTENCE MEANING

Expected value is the probability-weighted average outcome under a specified distribution. It is a long-run center, not the most likely result and not a guaranteed realization.

MEMORY JOGGER

Multiply each outcome by its weight and aggregate. For continuous variables, replace the weighted sum with an integral.

WHY IT MATTERS IN QUANT RESEARCH

Expected return, cost, slippage, payoff, and utility allow uncertain alternatives to be compared on one scale. The result is meaningful only when the relevant expectation exists and the model is credible.

FORMULA INTUITION

$E[X] = \sum x \times p(x)$ or $\int x f(x) dx$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A trade pays +2 units with probability 0.45 and -1 unit with probability 0.55. Its gross expectation is 0.35 units, but costs and selection effects may erase that edge.

PYTHON / SYSTEM IMPLEMENTATION

Compute analytic moments when stable and cross-check with simulation. Retain component contributions by state so changes in expectation can be attributed to probabilities versus payoffs.

CONFUSIONS TO AVOID

The mean can sit in a low-density region of a bimodal distribution and can be dominated by rare extremes. A positive expectation does not imply tolerable drawdown or positive median outcome.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Expected values enter ranking and allocation only after costs, constraints, uncertainty, and tail controls. The decision layer should preserve the full payoff distribution alongside its mean.

RELATED CONCEPTS AND QUICK RECALL

Related: mean, utility, tail risk. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Variance and standard deviation

BEGINNER

ONE-SENTENCE MEANING

Variance measures expected squared distance from the mean, and standard deviation returns that dispersion to the variable's original unit. Both summarize spread but ignore direction and much distributional shape.

MEMORY JOGGER

Square deviations so positive and negative moves do not cancel. Standard deviation is the square root needed for direct unit interpretation.

WHY IT MATTERS IN QUANT RESEARCH

Volatility, forecast uncertainty, and portfolio risk often use variance-based summaries. They are tractable under aggregation but can be fragile for heavy tails and asymmetric losses.

FORMULA INTUITION

$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$; $\sigma = \sqrt{\text{Var}(X)}$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

For outcomes -1 and +2 with probabilities 0.4 and 0.6, $E[X] = 0.8$ and $E[X^2] = 2.8$. Variance is 2.16 and standard deviation is about 1.47.

PYTHON / SYSTEM IMPLEMENTATION

Use numerically stable two-pass or online algorithms and explicit degrees-of-freedom settings. For weighted data, document the weight interpretation before applying a variance formula.

CONFUSIONS TO AVOID

Standard deviation treats upside and downside equally and can be undefined for some heavy-tailed models. Annualizing with square-root time assumes conditions that financial returns often violate.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Variance estimates feed covariance matrices, volatility forecasts, and risk budgets. Model governance should pair them with tail, skew, and stability diagnostics.

RELATED CONCEPTS AND QUICK RECALL

Related: second moment, volatility, covariance. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Transformations of random variables

BEGINNER

ONE-SENTENCE MEANING

A transformation creates $Y=g(X)$, changing support, density shape, units, and sometimes ordering. Probability is preserved through the mapping, but densities require a Jacobian adjustment.

MEMORY JOGGER

Transform the outcome rule first, then derive the new distribution. Monotone maps make CDF reasoning especially direct.

WHY IT MATTERS IN QUANT RESEARCH

Log returns, standardized signals, volatility stabilizers, and rank transforms can improve modeling or interpretation. Each also changes what fitted parameters and errors mean.

FORMULA INTUITION

If $Y=g(X)$ is one-to-one, $f_Y(y)=f_X(g^{-1}(y))|dg^{-1}(y)/dy|$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If price ratio R is lognormal, $\log(R)$ is normal. Modeling log price ratios respects positivity when transformed back, but expected levels require the lognormal correction.

PYTHON / SYSTEM IMPLEMENTATION

Register transformation code, inverse, domain, fitted parameters, and unit tests for round trips. Fit preprocessing only on training data and apply the frozen transform out of sample.

CONFUSIONS TO AVOID

Ignoring the Jacobian gives an invalid likelihood. Applying a log to zero or negative values, or leaking full-sample scaling statistics, creates silent model defects.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Transformations belong in versioned pipelines, not notebook-only preprocessing. Scenario outputs should be inverse-transformed before economic interpretation and limit checks.

RELATED CONCEPTS AND QUICK RECALL

Related: Jacobian, log transform, standardization. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Joint, marginal, and conditional distributions

BEGINNER

ONE-SENTENCE MEANING

A joint distribution describes variables together, marginals summarize one variable after integrating out others, and conditionals describe one variable given another. These are different views of one probability model.

MEMORY JOGGER

Joint is the whole surface; marginal is a projection; conditional is a normalized slice. Dependence lives in how the joint differs from a product of marginals.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio co-movement, regime-conditioned returns, feature-target relationships, and multistage simulations all require joint structure. Marginal fits alone cannot determine aggregate risk.

FORMULA INTUITION

$f_X(x) = \int f_{XY}(x,y) dy$; $f_{X|Y}(x|y) = f_{XY}(x,y) / f_Y(y)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A return-volatility joint model can show the unconditional return distribution and the much wider return distribution conditional on stressed volatility. The marginal hides that regime contrast.

PYTHON / SYSTEM IMPLEMENTATION

Store tidy joint samples and derive views from the same time-aligned dataset. Validate marginals recovered from the joint and conditionals normalized at representative conditioning values.

CONFUSIONS TO AVOID

Combining separately fitted marginals as if independent invents a joint model. Conditioning on a rare value can also produce unstable estimates with little local support.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dependency layer joins asset, factor, and regime marginals into portfolio scenarios. Its assumptions must be explicit inputs to risk and allocation services.

RELATED CONCEPTS AND QUICK RECALL

Related: dependence, copula, regime conditioning. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Bernoulli distribution

BEGINNER

ONE-SENTENCE MEANING

A Bernoulli variable records one binary outcome with success probability p . Its simplicity makes event definition and calibration more important, not less.

MEMORY JOGGER

One trial, two states, one probability parameter. Every binary label is Bernoulli only after the eligible population and success rule are fixed.

WHY IT MATTERS IN QUANT RESEARCH

Direction labels, limit breaches, fills, defaults, and signal activations are common Bernoulli outcomes. Their probabilities feed classification, event risk, and decision thresholds.

FORMULA INTUITION

$P(X=x)=p^x(1-p)^{(1-x)}$; $E[X]=p$; $\text{Var}(X)=p(1-p)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If next-day loss probability is 0.30, encode loss as 1 and no loss as 0. The expected indicator is 0.30 and its variance is 0.21.

PYTHON / SYSTEM IMPLEMENTATION

Use boolean or integer storage with explicit null handling. Evaluate log loss, Brier score, calibration, prevalence drift, and support by time and regime.

CONFUSIONS TO AVOID

Changing which class is coded as one reverses interpretation. Balancing training classes also changes the effective prior, so raw fitted scores may no longer be production probabilities.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The Bernoulli model is the atomic event contract for alerting, classifiers, and breach monitoring. The forecast ledger must store probabilities before outcomes are known.

RELATED CONCEPTS AND QUICK RECALL

Related: binary event, binomial, calibration. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Binomial distribution

BEGINNER

ONE-SENTENCE MEANING

A binomial variable counts successes across n trials that share success probability p and are independent conditional on the model. Both equal- p and independence assumptions are substantive.

MEMORY JOGGER

Repeat the same Bernoulli experiment n times and count successes. The count ranges from zero through n .

WHY IT MATTERS IN QUANT RESEARCH

Binomial models benchmark hit counts, defect counts, and event frequencies over fixed windows. They also provide exact uncertainty calculations when assumptions are defensible.

FORMULA INTUITION

$P(X=k)=C(n,k)p^k(1-p)^{(n-k)}$; $E[X]=np$; $\text{Var}(X)=np(1-p)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

With $n=20$ trades and $p=0.55$, expected wins are 11 and standard deviation is $\sqrt{4.95}=2.22$. Clustering across one regime makes that binomial spread too narrow.

PYTHON / SYSTEM IMPLEMENTATION

Use stable log-PMF functions for large n and exact survival functions for tail tests. Compare observed count dispersion across windows with $np(1-p)$.

CONFUSIONS TO AVOID

Serially dependent trades, varying probabilities, and overlapping labels violate the ordinary binomial model. A significant count can reflect a changed opportunity set rather than skill.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The binomial serves as a transparent baseline in experiment reports. When assumptions fail, route to beta-binomial, Poisson-binomial, or block-resampling alternatives.

RELATED CONCEPTS AND QUICK RECALL

Related: Bernoulli, beta-binomial, overdispersion. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Geometric and negative-binomial distributions

BEGINNER

ONE-SENTENCE MEANING

The geometric distribution models trials until the first success, while the negative binomial extends this to waiting for multiple successes or modeling overdispersed counts. Parameter conventions vary across libraries.

MEMORY JOGGER

Geometric is waiting for one event; negative binomial allows clustered counts or waiting for r events. Always state whether the count includes the success trial.

WHY IT MATTERS IN QUANT RESEARCH

They can model time between signals, attempts until fill, clustered trade arrivals, and count data whose variance exceeds its mean. Memorylessness applies only to the geometric case under constant p .

FORMULA INTUITION

$P(T=t)=(1-p)^{t-1}p$; $E[T]=1/p$ for $t=1,2,\dots$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If an order has independent 20% fill chance each second, expected waiting time is five seconds. Real fill probability usually changes with queue position and market state, breaking the model.

PYTHON / SYSTEM IMPLEMENTATION

Pin the library's parameterization in tests using known means and PMFs. For count regression, verify whether the dispersion parameter controls variance as $\mu+\mu^2/r$ or another convention.

CONFUSIONS TO AVOID

Off-by-one support errors are common and materially change likelihoods. Treating changing hazard or serial dependence as geometric behavior produces misleading tail waiting times.

WHERE THIS FITS IN THE RESEARCH SYSTEM

These models belong in execution and event-arrival modules with explicit clock, eligibility, and censoring fields. Diagnostics should compare observed and simulated waiting-time tails.

RELATED CONCEPTS AND QUICK RECALL

Related: waiting time, overdispersion, hazard. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Poisson distribution

BEGINNER

ONE-SENTENCE MEANING

A Poisson variable counts events in a fixed exposure when arrivals occur at a constant rate and independently under the model. Its mean and variance both equal the exposure-scaled rate.

MEMORY JOGGER

Count events over a defined interval, area, or volume. Rate is meaningless unless the exposure unit is attached.

WHY IT MATTERS IN QUANT RESEARCH

Poisson models provide baselines for trade arrivals, quote changes, outages, and rare operational events. Deviations reveal clustering, seasonality, or changing intensity.

FORMULA INTUITION

$P(X=k)=\exp(-\lambda)\lambda^k/k!$; $E[X]=\text{Var}(X)=\lambda$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If quote updates average 12 per second under a stable interval model, the benchmark count variance is also 12. A variance of 45 signals overdispersion or rate variation.

PYTHON / SYSTEM IMPLEMENTATION

Calculate in log space for large k and include exposure offsets in regression. Segment intraday seasonality before judging dispersion and inspect zero frequency separately.

CONFUSIONS TO AVOID

Market events are rarely homogeneous through time, and common shocks violate independent increments. Using one daily rate can hide opening, news, and closing intensity patterns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The arrival-rate service provides baseline intensity and anomaly scores to execution monitoring. Production models should include seasonality and regime challengers.

RELATED CONCEPTS AND QUICK RECALL

Related: count process, exponential waiting time, intensity. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Discrete uniform distribution

BEGINNER

ONE-SENTENCE MEANING

A discrete uniform distribution assigns equal probability to every value in a finite support. The symmetry must come from the experiment design rather than convenience.

MEMORY JOGGER

Every listed state gets the same mass. This is a defensible model for randomized selection, not a default model for market outcomes.

WHY IT MATTERS IN QUANT RESEARCH

It appears in permutation tests, random baselines, randomized tie-breaking, and controlled scenario sampling. It is valuable precisely because the randomization mechanism can be audited.

FORMULA INTUITION

$P(X=x)=1/m$ for x in $\{x_1, \dots, x_m\}$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Selecting one of 50 eligible assets with a verified random-number generator gives each asset probability $1/50$. Selecting from current survivors does not make historical assets equally eligible.

PYTHON / SYSTEM IMPLEMENTATION

Freeze and sort the eligible set, seed the generator, and persist the selected index plus set hash. Test approximate frequencies across many draws and exact endpoint coverage.

CONFUSIONS TO AVOID

Modulo-based integer generation can introduce bias when the random range is not divisible by m . Unstable ordering can also make the same seed select different assets.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Randomized controls should be registered beside research experiments so their population, seed, and purpose are reproducible. They provide a clean benchmark against selection effects.

RELATED CONCEPTS AND QUICK RECALL

Related: randomization, permutation, categorical. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Continuous uniform distribution

BEGINNER

ONE-SENTENCE MEANING

A continuous uniform distribution has constant density across a bounded interval and zero density outside it. Equal-length subintervals receive equal probability.

MEMORY JOGGER

Flat density does not mean equal probability for every point; each point still has probability zero. It means probability is proportional to interval length.

WHY IT MATTERS IN QUANT RESEARCH

Uniform draws power inverse-transform sampling, randomized parameters, and simple bounded stress tests. They are rarely realistic direct models of financial returns.

FORMULA INTUITION

$f(x)=1/(b-a)$ for $a \leq x \leq b$; $E[X]=(a+b)/2$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A $U(0,1)$ draw of 0.93 can be transformed through a target inverse CDF to generate a high-quantile scenario. The uniform number itself is not the market outcome.

PYTHON / SYSTEM IMPLEMENTATION

Use half-open interval conventions consistently and clamp only for documented numerical endpoints. Test the generator with moment, histogram, and serial-correlation diagnostics.

CONFUSIONS TO AVOID

Sampling parameters uniformly can overweight regions with little economic meaning, especially across orders of magnitude. A log-uniform or domain-informed prior may be more defensible.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The random-number layer supplies uniform variates to all simulation modules. Seeds, streams, generators, and transformations must be versioned for replay.

RELATED CONCEPTS AND QUICK RECALL

Related: inverse transform, random generator, bounded support. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Normal distribution

BEGINNER

ONE-SENTENCE MEANING

The normal distribution is a symmetric bell-shaped model determined by mean and variance. It is analytically convenient but does not automatically describe financial returns, especially in tails.

MEMORY JOGGER

Standardize by subtracting the mean and dividing by standard deviation. The resulting z-score locates an outcome on a common scale under the model.

WHY IT MATTERS IN QUANT RESEARCH

Normal models support measurement noise, aggregated errors, regression assumptions, and benchmark risk calculations. They are best used as explicit approximations with tail checks.

FORMULA INTUITION

$f(x) = \exp(-(x-\mu)^2/(2\sigma^2))/(\sigma \sqrt{2\pi})$; $Z=(X-\mu)/\sigma$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Under $N(0,1\%^2)$, a -3% return is $z=-3$ and has one-sided probability near 0.135%. Empirical equity tails usually make such moves far more common.

PYTHON / SYSTEM IMPLEMENTATION

Fit location and scale only on training windows, use stable logpdf/cdf/sf routines, and compare standardized residuals with Q-Q plots and exceedance counts.

CONFUSIONS TO AVOID

A visually good center fit can coexist with severe tail underestimation. Square-root-of-time scaling and independence assumptions may compound that error across horizons.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Normal models are useful baselines in the registry, not privileged truths. Risk engines should retain empirical and heavy-tail challengers for loss thresholds.

RELATED CONCEPTS AND QUICK RECALL

Related: z-score, CLT, Q-Q plot. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Lognormal distribution

BEGINNER

ONE-SENTENCE MEANING

A positive variable is lognormal when its logarithm is normally distributed. The model is right-skewed, bounded below by zero, and has a mean above its median.

MEMORY JOGGER

Normal in log space becomes multiplicative and positive in level space. Back-transforming a mean requires a variance correction.

WHY IT MATTERS IN QUANT RESEARCH

Lognormal models can approximate price relatives, positive sizes, and multiplicative growth over restricted horizons. They do not permit negative values or point mass at zero.

FORMULA INTUITION

If $\ln X \sim N(\mu, \sigma^2)$, $\text{median} = e^\mu$ and $E[X] = e^{(\mu + \sigma^2/2)}$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If log price ratio has $\mu=0$ and $\sigma=0.20$, median ratio is 1 but mean ratio is $\exp(0.02)=1.0202$. Volatility shifts the level-space mean upward.

PYTHON / SYSTEM IMPLEMENTATION

Estimate in log space after domain validation, then report both log and level parameters. Use log-sum-exp techniques and unit tests for moments and quantiles.

CONFUSIONS TO AVOID

Exponentiating the fitted log mean gives the median, not the arithmetic mean. Applying the model to quantities that can be zero or negative is structurally invalid.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Transformation metadata must accompany the fitted distribution through forecasting and simulation. Economic consumers should receive level-space scenarios with clear units.

RELATED CONCEPTS AND QUICK RECALL

Related: multiplicative process, geometric Brownian motion, Jensen's inequality. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Exponential distribution

BEGINNER

ONE-SENTENCE MEANING

The exponential distribution models positive waiting time under a constant hazard rate. It is memoryless, so elapsed waiting does not alter the modeled remaining-time distribution.

MEMORY JOGGER

Constant event intensity implies exponential gaps. Its mean waiting time is the reciprocal of the rate.

WHY IT MATTERS IN QUANT RESEARCH

It provides a baseline for time-to-fill, arrival gaps, component lifetimes, and outage timing. Departures from constant hazard reveal queue, age, or regime effects.

FORMULA INTUITION

$f(t) = \lambda e^{-\lambda t}$; $S(t) = e^{-\lambda t}$; $E[T] = 1/\lambda$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

At rate 0.25 fills per second, expected wait is four seconds and $P(T > 8) = e^{-2} = 13.5\%$. Opening-auction conditions may have a very different rate.

PYTHON / SYSTEM IMPLEMENTATION

Retain censored durations, fit exposure-consistent rates, and compare empirical survival on a log scale. Test memorylessness by conditioning remaining time on elapsed time.

CONFUSIONS TO AVOID

Estimating from completed waits only discards slow or unfilled orders and biases the rate upward. Mixing regimes creates a nonconstant aggregate hazard.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Execution analytics can use exponential behavior as a diagnostic baseline. More flexible duration models should replace it when residual hazard varies systematically.

RELATED CONCEPTS AND QUICK RECALL

Related: Poisson process, hazard, survival. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Gamma distribution

BEGINNER

ONE-SENTENCE MEANING

The gamma distribution models positive continuous outcomes with flexible right skew. It can represent waiting time to multiple Poisson events and contains the exponential as a special case.

MEMORY JOGGER

Shape controls skew and concentration; scale controls units. Higher shape makes the distribution less sharply right-skewed.

WHY IT MATTERS IN QUANT RESEARCH

Gamma models are useful for positive costs, realized variance, durations, and aggregate waiting times. They preserve positivity while allowing more shape flexibility than the exponential.

FORMULA INTUITION

$f(x) = x^{(k-1)} e^{(-x/\theta)} / (\Gamma(k) \theta^k)$; $E[X] = k \theta$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

With shape $k=4$ and scale $\theta=0.5$ seconds, mean duration is two seconds and variance is one second squared. Different pairs can share the same mean but have different spread.

PYTHON / SYSTEM IMPLEMENTATION

Lock shape-rate versus shape-scale conventions with reference tests. Fit via stable log-likelihood and inspect lower-bound behavior plus upper-tail quantiles.

CONFUSIONS TO AVOID

Confusing rate β with scale θ inverts the parameter and can silently produce huge errors. A gamma fit also cannot represent exact zeros without an added point mass.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Positive-outcome models should expose parameter conventions in metadata and API names. Scenario tests should compare analytic and simulated moments.

RELATED CONCEPTS AND QUICK RECALL

Related: exponential, chi-square, positive support. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Beta distribution

BEGINNER

ONE-SENTENCE MEANING

The beta distribution models continuous values between zero and one with two positive shape parameters. It can be symmetric, skewed, U-shaped, or concentrated near an interior mean.

MEMORY JOGGER

Beta is a flexible distribution for proportions, not binary outcomes themselves. Boundary values may require special treatment because ordinary beta support is open.

WHY IT MATTERS IN QUANT RESEARCH

It can model uncertain rates, bounded signal strengths, recovery fractions, and cross-sectional proportions. In Bayesian work it is conjugate to the Bernoulli likelihood.

FORMULA INTUITION

$f(x) = x^{a-1}(1-x)^{b-1}/B(a,b)$; $E[X] = a/(a+b)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A Beta(8,2) distribution has mean 0.8 but still represents uncertainty around that rate. It is not the same as observing a fixed 80% success probability with certainty.

PYTHON / SYSTEM IMPLEMENTATION

Use constrained optimization or mean-precision parameterization, and decide how exact zeros and ones are handled. Validate calibration by comparing predicted beta quantiles with observed proportions.

CONFUSIONS TO AVOID

Adding a tiny epsilon to boundaries changes the data-generating story and can bias shape estimates. A beta distribution also cannot model point mass at zero or one without inflation components.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Bounded-parameter distributions belong in forecast-uncertainty and hierarchical-rate modules. Their evidence counts and prior assumptions must remain visible.

RELATED CONCEPTS AND QUICK RECALL

Related: proportion, beta-binomial, Bayesian prior. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Student t distribution

BEGINNER

ONE-SENTENCE MEANING

The Student t distribution is symmetric like the normal but has heavier tails controlled by degrees of freedom. It converges toward the normal as degrees of freedom increase.

MEMORY JOGGER

Low degrees of freedom mean more extreme observations and less stable high moments. Location-scale t parameters are not always identical to mean and standard deviation.

WHY IT MATTERS IN QUANT RESEARCH

The t distribution often provides a stronger return or residual baseline than the normal. It can absorb tail thickness but does not model skew or volatility clustering by itself.

FORMULA INTUITION

$f(z)$ proportional to $(1+z^2/\nu)^{-(\nu+1)/2}$; Var exists only for $\nu > 2$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A fitted t with $\nu=4$ assigns far more probability to large standardized losses than a normal. Its fourth moment is infinite, so kurtosis-based summaries become problematic.

PYTHON / SYSTEM IMPLEMENTATION

Fit location, scale, and degrees of freedom with bounded parameters, then inspect profile likelihood and tail exceedances. Report whether requested moments mathematically exist.

CONFUSIONS TO AVOID

Calling the scale parameter standard deviation is wrong when variance depends on ν . Heavy tails can also be a symptom of regime mixtures rather than a single stationary t process.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The distribution registry should store existence checks for moments and compare t, normal, empirical, and regime-mixture challengers on held-out tails.

RELATED CONCEPTS AND QUICK RECALL

Related: heavy tail, degrees of freedom, robust residual. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Chi-square and F distributions

BEGINNER

ONE-SENTENCE MEANING

Chi-square distributions arise from sums of squared standard normals, while F distributions arise from ratios of scaled chi-square variables. They support classical variance and nested-model inference under restrictive assumptions.

MEMORY JOGGER

Chi-square is squared Gaussian energy; F compares two variance-like quantities. Degrees of freedom determine their skew and reference shape.

WHY IT MATTERS IN QUANT RESEARCH

They appear in variance confidence intervals, regression tests, likelihood diagnostics, and model comparisons. Financial dependence and nonnormality can invalidate textbook reference distributions.

FORMULA INTUITION

If $Z_i \sim N(0,1)$, $\sum Z_i^2 \sim \text{chi-square}(k)$; $(U/d1)/(V/d2) \sim F(d1,d2)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Under independent Gaussian residuals, a residual sum-of-squares ratio can follow an F reference law. Autocorrelated or heteroskedastic errors alter that benchmark.

PYTHON / SYSTEM IMPLEMENTATION

Use library survival functions for p-values, state numerator and denominator degrees of freedom, and validate assumptions with residual diagnostics or robust alternatives.

CONFUSIONS TO AVOID

A small p-value does not measure economic importance, and repeated testing inflates false discoveries. Blindly using iid reference laws on time-series residuals gives spurious certainty.

WHERE THIS FITS IN THE RESEARCH SYSTEM

These reference distributions belong in the inference layer with assumption flags. Experiment reports should pair them with effect sizes, robust errors, and out-of-sample evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: variance test, regression, degrees of freedom. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Mixture distributions

BEGINNER

ONE-SENTENCE MEANING

A mixture distribution combines component distributions using nonnegative weights that sum to one. It can represent latent regimes, subpopulations, or multiple mechanisms producing the same observed variable.

MEMORY JOGGER

Choose a component according to its weight, then draw from that component. The aggregate may be skewed, heavy-tailed, or multimodal even when components are simple.

WHY IT MATTERS IN QUANT RESEARCH

Mixtures model calm and stressed returns, liquid and illiquid execution, or several asset groups. They separate changing regime frequency from changing within-regime behavior.

FORMULA INTUITION

$f(x) = \sum_j \pi_j f_j(x)$; $\pi_j \geq 0$; $\sum_j \pi_j = 1$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

An 85% $N(0, 1\%^2)$ calm component plus 15% $N(-1\%, 3\%^2)$ stress component has a loss-heavy aggregate tail. A single normal blurs both states.

PYTHON / SYSTEM IMPLEMENTATION

Initialize fits multiple ways, constrain degenerate scales, and retain posterior component probabilities. Use time-split checks because label switching and local optima complicate comparison.

CONFUSIONS TO AVOID

Components need not correspond to real economic regimes, and flexible mixtures can overfit noise. Interpreting labels without stability and external evidence is hazardous.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The regime layer can publish component weights and state-conditioned scenarios to forecasting and risk. Governance should track stability, transition logic, and a simpler challenger.

RELATED CONCEPTS AND QUICK RECALL

Related: latent state, regime, multimodality. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Empirical distributions and ECDFs

BEGINNER

ONE-SENTENCE MEANING

An empirical distribution assigns equal mass to observed sample values, and its ECDF records the fraction at or below each threshold. It makes few shape assumptions but inherits every sample bias.

MEMORY JOGGER

Let the data speak through observed points. The ECDF is a staircase whose jumps reveal finite support and ties.

WHY IT MATTERS IN QUANT RESEARCH

Empirical distributions provide transparent baselines for returns, residuals, slippage, and stress scenarios. They preserve observed asymmetry and tails within the historical sample.

FORMULA INTUITION

$F_n(x) = (1/n) \sum_i I(x_i \leq x)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If 12 of 500 returns are at or below -3%, the empirical CDF there is 2.4%. The estimate says nothing reliable about losses beyond the worst observed value.

PYTHON / SYSTEM IMPLEMENTATION

Store sorted values, weights, dates, and cohort metadata. Use weighted ECDFs only when the weights have a documented sampling or economic interpretation.

CONFUSIONS TO AVOID

Historical observations are not necessarily iid or representative of the future. Resampling individual days also destroys serial dependence and path structure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The empirical model is a mandatory challenger in the distribution registry. Block bootstrap and regime-stratified variants should be available for scenario generation.

RELATED CONCEPTS AND QUICK RECALL

Related: histogram, bootstrap, nonparametric. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Histograms and kernel density estimates

BEGINNER

ONE-SENTENCE MEANING

A histogram groups observations into bins, while a kernel density estimate smooths them into a continuous curve. Both are visualization and estimation choices whose tuning can change the apparent shape.

MEMORY JOGGER

Bins reveal counts; kernels reveal a smoothed contour. Neither removes sampling noise or repairs biased data.

WHY IT MATTERS IN QUANT RESEARCH

These tools expose skew, multiple modes, outliers, and scale differences before formal fitting. Comparing them across regimes can reveal instability hidden by aggregate statistics.

FORMULA INTUITION

Histogram density = $\text{count}/(n \cdot \text{bin width})$; KDE = $(1/(nh)) \sum K((x-x_i)/h)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A wide bandwidth can merge calm and stressed return modes into one bell, while a narrow bandwidth can turn noise into false peaks. Both views should be checked.

PYTHON / SYSTEM IMPLEMENTATION

Choose bins with a documented rule, show raw counts, and run bandwidth sensitivity. Overlay candidate PDFs only after matching units and normalization.

CONFUSIONS TO AVOID

Using the visual that best supports a hypothesis is researcher discretion. Boundary bias can also create density outside feasible domains such as negative spreads.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploratory diagnostics should save parameters and data hashes beside plots. Automated reports can show several defensible resolutions rather than one curated image.

RELATED CONCEPTS AND QUICK RECALL

Related: density estimate, bandwidth, visualization. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Sums and linear combinations

BEGINNER

ONE-SENTENCE MEANING

A linear combination of random variables creates a new variable whose mean is the same combination of means. Its variance additionally depends on every covariance pair.

MEMORY JOGGER

Means add directly; risks interact through dependence. Portfolio weights transform a joint return distribution into a P&L distribution.

WHY IT MATTERS IN QUANT RESEARCH

This is the mathematical core of portfolio return, factor exposure, spread construction, and aggregated cost. Marginal distributions alone are insufficient for the result.

FORMULA INTUITION

$Y = a^T X$; $E[Y] = a^T \mu$; $\text{Var}(Y) = a^T \Sigma a$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

For two equal-volatility assets, a 50/50 portfolio has much lower variance when correlation is zero than when correlation is 0.9. Expected return is unchanged by that correlation.

PYTHON / SYSTEM IMPLEMENTATION

Align timestamps and units, estimate covariance on the same sample, and validate matrix symmetry and positive semidefiniteness. Compare analytic moments with simulated portfolio outcomes.

CONFUSIONS TO AVOID

Assuming independence removes covariance terms and typically understates common-shock risk. Applying weights to mismatched return horizons creates a quantity with no coherent meaning.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The portfolio engine consumes joint scenarios or covariance models plus versioned weights. Risk attribution should preserve marginal and cross-term contributions.

RELATED CONCEPTS AND QUICK RECALL

Related: covariance, convolution, portfolio P&L. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Order statistics, minima, and maxima

BEGINNER

ONE-SENTENCE MEANING

Order statistics are sorted sample values; minima and maxima describe extremes whose distributions differ sharply from that of an ordinary draw. Their behavior depends strongly on sample size.

MEMORY JOGGER

The more draws you take, the more extreme the maximum and minimum tend to become. Selection changes the reference distribution.

WHY IT MATTERS IN QUANT RESEARCH

Cross-sectional winners, worst drawdowns, top-ranked signals, and largest slippage observations are selected extremes. Comparing them with ordinary observations creates selection bias.

FORMULA INTUITION

For iid draws, $P(\max X_i \leq x) = F(x)^n$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

If one asset exceeds a threshold with probability 1%, the chance at least one of 100 independent assets exceeds it is about 63.4%, not 1%.

PYTHON / SYSTEM IMPLEMENTATION

Carry the candidate count and selection rule into every ranked artifact. Use multiple-testing, bootstrap, or held-out replication to evaluate selected extremes.

CONFUSIONS TO AVOID

Treating the best backtest among hundreds as an ordinary backtest ignores search. Correlated candidates reduce but do not eliminate the selection effect.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Experiment tracking should record the entire search space, not only the winner. Validation allocates fresh data to the selected candidate after ranking is complete.

RELATED CONCEPTS AND QUICK RECALL

Related: multiple testing, extreme value, ranking. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Truncation and censoring

BEGINNER

ONE-SENTENCE MEANING

Truncation removes observations outside a rule from the sample, while censoring retains the case but records only that its value lies beyond a limit. They require different likelihoods and interpretations.

MEMORY JOGGER

Truncated values disappear; censored values remain with partial information. Both alter the observed distribution relative to the latent one.

WHY IT MATTERS IN QUANT RESEARCH

Price limits, reporting caps, unfilled orders, delistings, and data-vendor filters create these patterns. Ignoring them biases means, tails, duration estimates, and event rates.

FORMULA INTUITION

Right-censored case contributes $S(c)$; truncated density uses $f(x)/(F(b)-F(a))$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

An order still unfilled when monitoring ends is right-censored, not a fast zero-duration fill and not a row to drop. A vendor deleting returns beyond 50% creates truncation.

PYTHON / SYSTEM IMPLEMENTATION

Store censoring indicators, bounds, and reason codes. Fit censored or truncated likelihoods and verify recovery on synthetic datasets with known parameters.

CONFUSIONS TO AVOID

Winsorization is neither censoring nor a harmless cleanup; it replaces values and creates artificial boundary mass. Dropping censored cases systematically favors short durations.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Data-quality contracts must identify these mechanisms before modeling. Distribution services should reject ordinary fits when censoring or truncation flags are present but unsupported.

RELATED CONCEPTS AND QUICK RECALL

Related: missingness, survival analysis, winsorization. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Zero inflation and point-mass mixtures

BEGINNER

ONE-SENTENCE MEANING

Zero-inflated models combine a structural point mass at zero with a separate distribution for nonzero outcomes. They distinguish no-opportunity states from small continuous or count outcomes.

MEMORY JOGGER

One mechanism decides whether the outcome is exactly zero; another governs its size when active. This is not the same as a single distribution with many near-zero values.

WHY IT MATTERS IN QUANT RESEARCH

Trading volume, returns at illiquid timestamps, slippage, claims, and event counts may contain excess zeros. Separating occurrence from magnitude improves interpretation and forecasts.

FORMULA INTUITION

$P(X=0)=\pi+(1-\pi)P_{\text{base}}(0)$; nonzero mass scaled by $1-\pi$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A strategy trades on 8% of days and earns a continuous P&L only when active. Modeling all days with one Gaussian confounds participation probability with conditional payoff.

PYTHON / SYSTEM IMPLEMENTATION

Preserve exact zeros, measurement precision, and activity flags. Fit hurdle and zero-inflated alternatives, then compare occurrence calibration and conditional magnitude separately.

CONFUSIONS TO AVOID

Some zeros are missing values, stale prices, or rounding artifacts rather than structural outcomes. A flexible zero model can conceal a data-quality problem.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Signal generation should publish activation probability separately from conditional payoff distribution. Allocation can then model turnover, capacity, and expected P&L coherently.

RELATED CONCEPTS AND QUICK RECALL

Related: hurdle model, mixture, participation. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Sampling from distributions

BEGINNER

ONE-SENTENCE MEANING

Sampling converts a model and random-number stream into synthetic realizations. Correct draws require the intended distribution, dependence structure, parameter state, and reproducible transformation.

MEMORY JOGGER

Uniform random bits are transformed into modeled outcomes. A correct marginal sampler can still produce a wrong joint path.

WHY IT MATTERS IN QUANT RESEARCH

Simulation supports pricing, stress testing, uncertainty propagation, path-dependent backtests, and analytic verification. Sampling error is distinct from parameter and model error.

FORMULA INTUITION

Inverse transform: $X = F^{-1}(U)$, $U \sim \text{Uniform}(0,1)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

Drawing independent t returns matches the one-day marginal but misses volatility clustering. A path-level simulator must also reproduce conditional scale and dependence through time.

PYTHON / SYSTEM IMPLEMENTATION

Use independent named streams, fixed seeds for tests, vectorized batches, and deterministic replay metadata. Validate support, moments, quantiles, dependence, and path statistics.

CONFUSIONS TO AVOID

Reusing seeds across supposedly independent components can create hidden correlation. Increasing scenario count only reduces Monte Carlo noise around the chosen model.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scenario platform owns generators, stream allocation, model versions, and convergence evidence. Consumers should receive scenario identifiers so any decision can be replayed.

RELATED CONCEPTS AND QUICK RECALL

Related: Monte Carlo, random seed, inverse CDF. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Distribution fitting and maximum likelihood

BEGINNER

ONE-SENTENCE MEANING

Distribution fitting selects parameters that make observed data plausible under a stated family; maximum likelihood maximizes the joint likelihood or summed log-likelihood. It does not prove the family is true.

MEMORY JOGGER

Choose the model family first, then estimate its parameters from eligible data. Log likelihood turns products into sums and improves numerical stability.

WHY IT MATTERS IN QUANT RESEARCH

Fitting produces forecast distributions, residual models, and scenario parameters. Honest evaluation requires time separation, parameter uncertainty, and challenger families.

FORMULA INTUITION

$\theta_{\text{hat}} = \arg\max_{\theta} \sum_i \log f(x_i | \theta)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A t distribution may achieve higher training likelihood than a normal by fitting outliers. Held-out tail calibration determines whether that flexibility generalizes.

PYTHON / SYSTEM IMPLEMENTATION

Use bounded parameterization, stable logpdf functions, multiple starts, convergence checks, and stored optimizer traces. Fit only on training windows and score untouched periods.

CONFUSIONS TO AVOID

Highest in-sample likelihood rewards complexity and can chase regime-specific noise. Optimization convergence also does not guarantee global optimum or valid standard errors.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment registry stores family, parameters, window, optimizer, code hash, diagnostics, and held-out scores. Promotion requires both statistical and economic validation.

RELATED CONCEPTS AND QUICK RECALL

Related: log likelihood, parameter estimation, model selection. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Goodness-of-fit diagnostics

BEGINNER

ONE-SENTENCE MEANING

Goodness-of-fit diagnostics compare observed behavior with what a fitted distribution predicts across the center, tails, and relevant conditionals. No single score certifies adequacy for every decision.

MEMORY JOGGER

Inspect where the model fails, not only one aggregate distance. A risk model needs strong loss-tail evidence even if its center is imperfect.

WHY IT MATTERS IN QUANT RESEARCH

Q-Q plots, PIT histograms, exceedance counts, residual tests, and held-out scores reveal different defects. Decision relevance determines which discrepancies matter most.

FORMULA INTUITION

$\text{PIT } u_i = F_{\hat{}}(x_i)$; under a correct continuous model, u_i should be $\text{Uniform}(0,1)$. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A nearly straight central Q-Q plot can bend sharply in the lower tail, showing that modeled crash probabilities are too small. Average log score may obscure that local failure.

PYTHON / SYSTEM IMPLEMENTATION

Generate diagnostics by time, asset, regime, and forecast horizon. Use bootstrap bands or simulation envelopes and preserve the fitted-window distinction.

CONFUSIONS TO AVOID

Testing on the same data used for fitting changes reference distributions and overstates performance. Repeatedly choosing models from diagnostic plots also creates selection bias.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Validation artifacts should map each diagnostic to a consuming decision and threshold. Tail failures can block risk use even when a model remains acceptable for central forecasting.

RELATED CONCEPTS AND QUICK RECALL

Related: Q-Q plot, PIT, calibration. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Tail behavior and moment existence

BEGINNER

ONE-SENTENCE MEANING

Tail behavior describes how quickly extreme-event probability decays, and it determines whether moments such as variance or kurtosis exist. Similar-looking central distributions can imply radically different risk.

MEMORY JOGGER

Ask how fast survival probability falls and which expectations are finite. Sample moments can look stable even when the model's corresponding moment is undefined.

WHY IT MATTERS IN QUANT RESEARCH

Tail assumptions drive crash probability, expected shortfall, capital needs, and convergence rates. They matter more as the decision threshold moves beyond observed support.

FORMULA INTUITION

Power tail $S(x)$ approximately $Cx^{-\alpha}$; $E[X^k]$ finite only when $\alpha > k$ under this form. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A tail index near 3 can permit finite variance but infinite fourth moment. Empirical kurtosis then behaves erratically as the sample grows.

PYTHON / SYSTEM IMPLEMENTATION

Estimate tails over sensitivity ranges, report exceedance counts, and compare empirical, t, generalized Pareto, and stress alternatives. Mark extrapolated probability levels explicitly.

CONFUSIONS TO AVOID

Choosing a threshold after seeing the answer understates uncertainty. Reporting many decimal places for a probability based on a handful of extremes creates false precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The tail-risk layer stores threshold, support, model family, parameter uncertainty, and stress override separately. Capital decisions should use conservative model ensembles when evidence is thin.

RELATED CONCEPTS AND QUICK RECALL

Related: heavy tail, extreme value, expected shortfall. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Choosing a distribution for research

BEGINNER

ONE-SENTENCE MEANING

Distribution choice is a structured modeling decision based on support, mechanism, dependence, tails, interpretability, and out-of-sample performance. Familiarity or visual resemblance alone is insufficient.

MEMORY JOGGER

Start with what values are possible and how they arise. Then compare the simplest defensible model with empirical and stress challengers.

WHY IT MATTERS IN QUANT RESEARCH

A good choice makes assumptions visible, produces decision-relevant probabilities, and fails detectably. The best central fit may not be the best model for sizing or risk limits.

FORMULA INTUITION

Decision loss = $\sum_s P(s) L(\text{action}, s)$; model choice should reflect the loss-sensitive region. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

For slippage, a zero-inflated positive model may reflect no-fill versus filled-cost mechanisms better than a Gaussian. For residual noise, a t model may be an adequate robust baseline.

PYTHON / SYSTEM IMPLEMENTATION

Use a checklist covering support, zeros, censoring, conditioning, dependence, sample size, tail need, estimation stability, and held-out scoring. Register champion and challengers.

CONFUSIONS TO AVOID

Trying many families and reporting only the winner leaks model-search discretion into results. Mechanistic plausibility cannot be replaced by a marginal goodness-of-fit statistic.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Distribution selection is governed at experiment promotion and revisited under drift. Consumers should know which regions are interpolated, extrapolated, or stress-defined.

RELATED CONCEPTS AND QUICK RECALL

Related: model risk, challenger, decision loss. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Distribution contracts in quant systems

BEGINNER

ONE-SENTENCE MEANING

A distribution contract is the machine-readable specification of a random variable, fitted model, conditioning state, and supported operations. It makes probability outputs reproducible across research and production.

MEMORY JOGGER

The object is more than parameters: it includes meaning, units, time, transformations, data lineage, and validation. Without that context, a number like $\sigma=0.02$ is ambiguous.

WHY IT MATTERS IN QUANT RESEARCH

Contracts prevent mismatched horizons, stale parameters, unsupported tails, and inconsistent sampling across services. They also make model review and rollback practical.

FORMULA INTUITION

Contract = variable schema + family + parameters + conditioning + lineage + validation. Read every symbol against the variable's support, unit, horizon, and conditioning state.

SIMPLE MARKET EXAMPLE

A one-day return t model must state decimal units, close convention, degrees of freedom, location, scale, fit window, asset universe, and whether scenarios are independent through time.

PYTHON / SYSTEM IMPLEMENTATION

Expose pdf/pmf, cdf, sf, quantile, sample, moments, and validation metadata through one versioned interface. Reject unsupported operations such as nonexistent variance.

CONFUSIONS TO AVOID

Serializing parameters without transformations or parameter conventions creates irreproducible results. Allowing silent fallbacks can turn model errors into apparently valid probabilities.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry publishes approved contracts to forecasting, risk, portfolio, and execution services. Runtime monitors compare observed support, PIT behavior, and drift with the registered assumptions.

RELATED CONCEPTS AND QUICK RECALL

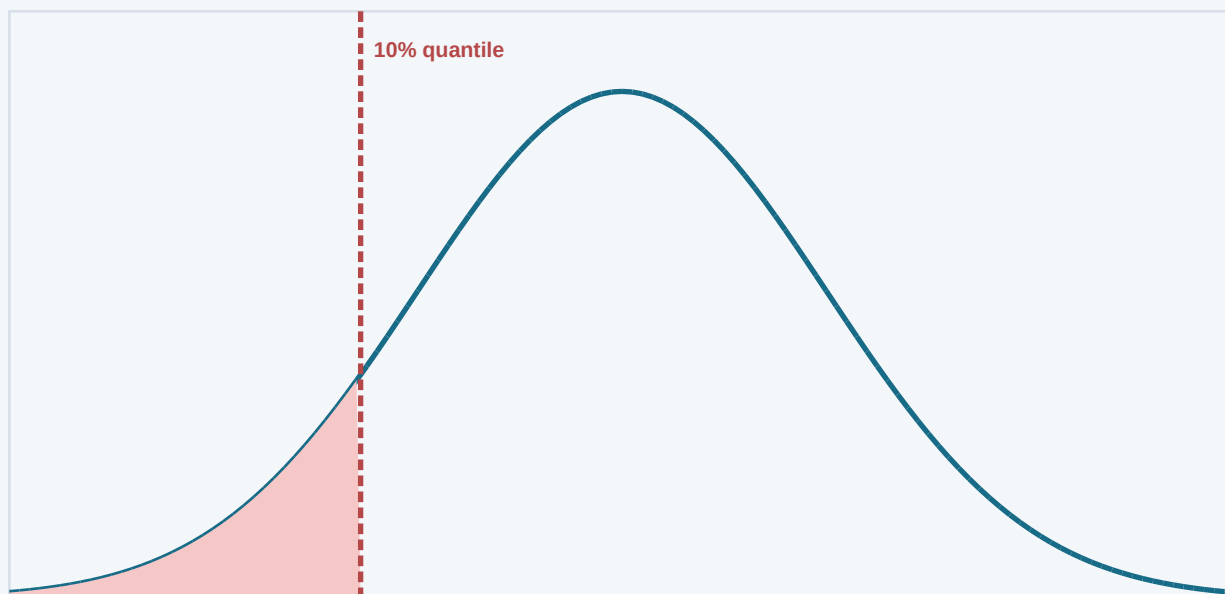
Related: model registry, lineage, validation gate. Reconstruct the support, the core probability rule, and one realistic failure mode before using this model in research.

Density, CDF, survival, and quantile

BEGINNER

FOUR VIEWS, FOUR QUESTIONS

Density shows local concentration; CDF shows mass below a threshold; survival shows mass above it; quantile returns the threshold for a chosen probability. They describe one distribution through different interfaces.



WORKED READING

For a standard normal, $Q(0.10)$ is about -1.28 . The shaded area is $F(-1.28)=10\%$; the opposite area is $S(-1.28)=90\%$. Density height at -1.28 is neither of those probabilities.

TAIL NUMERICS

In an extreme upper tail, compute $S(x)$ directly rather than $1-F(x)$, which can subtract nearly equal floating-point numbers. Use log-survival when the probability is too small for ordinary representation.

DISCRETE CONTRAST

For a discrete variable, the CDF jumps by $p(x)$ at support value x . A requested quantile can therefore map an interval of probability levels to one value; state the inverse-CDF convention.

QUICK RECALL

Given a loss limit, choose CDF or survival based on inequality direction. Given a percentile level, choose the quantile. Given an observed value for likelihood, choose PMF or density according to support.

Distribution family decision map

BEGINNER

START WITH MECHANISM AND SUPPORT

This map narrows candidates; it does not select a model automatically. Every branch still needs empirical, held-out, regime, and tail validation.

OUTCOME TYPE?

PATH A: Discrete. PATH B: Continuous.

DISCRETE MECHANISM

PATH A: Binary -> Bernoulli; fixed trials -> binomial. PATH B: Counts over exposure -> Poisson; overdispersed -> negative binomial.

CONTINUOUS SUPPORT

PATH A: Bounded $[0,1]$ -> beta; bounded interval -> uniform. PATH B: Positive -> exponential/gamma/lognormal; real line -> normal/t.

SPECIAL STRUCTURE

PATH A: Excess zeros -> hurdle or zero-inflated. PATH B: Multiple modes/regimes -> mixture; limits -> truncated/censored.

REQUIRED CHALLENGER

PATH A: Empirical PMF and resampling baseline. PATH B: ECDF or block-resampled empirical distribution.

PROMOTION QUESTIONS

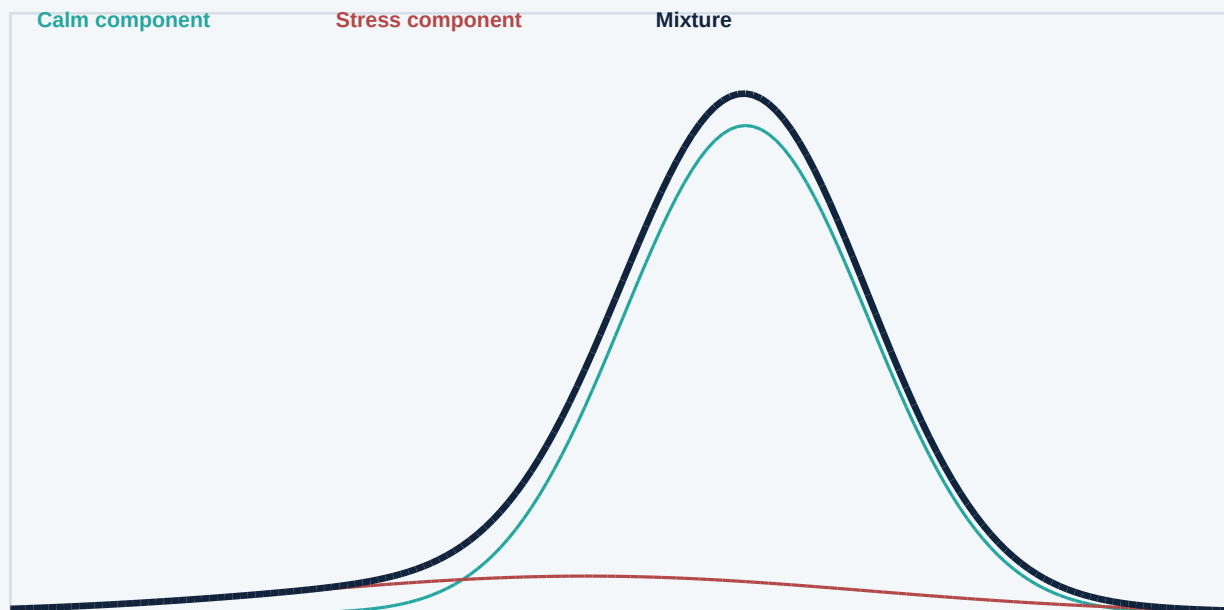
Does support match? Are zeros and boundaries real? Is dependence modeled? Which tail is decision-relevant? Are moments finite? Did the family beat a simple challenger on untouched data? Can production reproduce the transform and parameter convention?

Regime mixtures and hidden tail risk

BEGINNER

AGGREGATE SHAPE CAN HIDE MECHANISMS

A mixture first selects a latent state and then draws from its conditional distribution. Changing state weights can shift aggregate risk even when every component remains unchanged.



CONTRIBUTION ARITHMETIC

At threshold x , mixture loss probability is $0.85 F_{\text{calm}}(x) + 0.15 F_{\text{stress}}(x)$. The stress state can contribute most tail mass despite its small weight because its conditional distribution is wider and shifted left.

MONITORING DECOMPOSITION

When aggregate tail risk rises, decompose the change into regime-weight movement and within-regime parameter movement. This identifies whether the market is entering stress more often or stress itself is worsening.

INTERPRETATION WARNING

Statistical components are not automatically economic regimes. Require temporal stability, interpretable covariates, transition evidence, and improved held-out decisions before assigning labels such as calm or crisis.

SYSTEM FIT

Publish component weights, conditional scenarios, and aggregate scenarios together. This preserves attribution and lets risk consumers stress either state frequency or state severity.

Formula sheet and worked calculations

BEGINNER

DISCRETE EXPECTATION

For X in $\{-2, 0, 3\}$ with probabilities $\{0.20, 0.50, 0.30\}$, $E[X] = -0.4 + 0 + 0.9 = 0.5$. $E[X^2] = 0.8 + 0 + 2.7 = 3.5$, so $\text{Var}(X) = 3.5 - 0.25 = 3.25$.

BINOMIAL EVENT COUNT

For $n=40$ independent opportunities with $p=0.15$, expected events are 6 and variance is 5.1. Standard deviation is 2.26; dependence or varying p requires another model.

EXPONENTIAL WAITING TIME

At $\lambda=0.4$ events per second, mean wait is 2.5 seconds. Probability of waiting beyond five seconds is $\exp(-2)=13.5\%$, and constant hazard is the key assumption.

LOGNORMAL CORRECTION

If $\ln X \sim N(0.01, 0.20^2)$, median X is $\exp(0.01)=1.0101$ while mean X is $\exp(0.01+0.02)=1.0305$. Exponentiating the log mean alone misses Jensen's correction.

MIXTURE TAIL

With calm weight 0.9 and 1% conditional breach rate plus stress weight 0.1 and 20% breach rate, total breach probability is $0.009 + 0.020 = 2.9\%$. Stress supplies 69% of breach mass.

PORTFOLIO LINEAR COMBINATION

For weights a , mean is $a^T \mu$ and variance is $a^T \Sigma a$. Linearity of expectation does not need independence; variance aggregation needs covariance information.

Failure modes and diagnostic table

BEGINNER

IMPOSSIBLE SIMULATED VALUES

Audit support, transformations, parameter convention, and clipping. Replace the family or use a principled truncated model; do not hide the defect at the output boundary.

CENTER FITS, LOSSES DO NOT

Inspect lower-tail Q-Q behavior, exceedance counts, survival probabilities, regime conditioning, and empirical challengers. Block use for risk limits until the consumed tail is validated.

TOO MANY ZEROS

Separate structural inactivity, rounding, stale observations, and missing data. Compare hurdle, zero-inflated, and data-cleaning explanations before fitting.

VARIANCE EXCEEDS COUNT MEAN

A Poisson baseline is underdispersed. Check seasonality, common shocks, exposure heterogeneity, clustering, and negative-binomial or Cox-process alternatives.

PARAMETERS SHIFT ACROSS WINDOWS

Measure estimation uncertainty, support changes, regime weights, and population drift. Prefer conditional or hierarchical models when the shifts have stable drivers.

SIMULATION CONVERGES TO WRONG ANSWER

Monte Carlo convergence addresses sampling error only. Recheck family, parameters, dependence, transformations, payoff code, and analytic toy cases.

Implementation and model-registry checklist

BEGINNER

VARIABLE CONTRACT

Name, experiment, outcome map, support, unit, horizon, clock, population, missing policy, censoring, owner, and version.

EVIDENCE SNAPSHOT

Source tables, as-of timestamp, eligibility, row count, effective sample size, zeros, boundaries, exclusions, regimes, and immutable hash.

MODEL SPECIFICATION

Family, parameter convention, transformations, conditioning, fit window, objective, optimizer, initialization, random seed, and code version.

VALIDATION EVIDENCE

Held-out scores, ECDF comparison, PIT or Q-Q diagnostics, support tests, tail counts, stability, parameter uncertainty, and challenger results.

RUNTIME INTERFACE

pmf/pdf, logpmf/logpdf, cdf, sf, quantile, sample, moments, unsupported-operation errors, units, and deterministic replay.

MONITORING AND ROLLBACK

Support breaches, PIT drift, tail exceptions, parameter age, regime-weight movement, schema compatibility, approval, fallback, and rollback artifact.

Final active recall index

BEGINNER

FOUNDATIONS

Define random variable, realization, sample, population, support, discrete mass, continuous density, CDF, survival, and quantile using one market outcome.

CORE FAMILIES

For Bernoulli, binomial, geometric, Poisson, uniform, normal, lognormal, exponential, gamma, beta, and Student t, state support, parameters, one use, and one failure.

STRUCTURE

Explain transformations, Jacobians, joint versus marginal views, linear combinations, order statistics, mixtures, zero inflation, truncation, and censoring.

ESTIMATION

Contrast empirical distributions, histograms, KDE, maximum likelihood, Q-Q plots, PIT diagnostics, held-out scoring, and parameter uncertainty.

SIMULATION

Explain inverse-transform sampling, seeds, independent streams, dependence modeling, convergence, and why more scenarios cannot repair a misspecified model.

PROMOTION TEST

Before publishing a distribution, answer: what is X , what can it equal, how was it sampled, which mechanism supports the family, where was fit validated, which tail matters, and how can production replay it?